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ORTHOPEDIC IMPLANT AND RELATED ELECTRODE ASSEMBLY

Abstract

Various implementations include orthopedic implants and related electrode assemblies. In a particular implementation, an electrode assembly includes: a first housing containing at least one electrode; and a control module with a first connector for mating with a complementary connector on the first housing, the control module including a controller programmed to initiate transmission of at least one of a measurement pattern or a stimulation pattern to a patient with the at least one electrode, wherein the first housing and the control module include a biocompatible material, and when connected, are sized to complement an orthopedic implant.

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Background/Summary

TECHNICAL FIELD

[0001] This disclosure generally relates to the field of surgery and surgical implants. More particularly, the disclosure relates to implants, assemblies, and related systems enabling treatment and/or monitoring of post-operative patient progress.

BACKGROUND

[0002] A promising frontier in orthopedics is the conception and commercialization of smart implants. In the context of orthopedics, the term “smart implant” describes a traditional orthopedic implant with enhanced functionality enabled by integrated sensing, processing, communication, power, actuation, and/or treatment functions. Conventional sensing and stimulation functions for smart implants involve the placement of one or more conductive elements (i.e. electrodes) at the site of interest. However, conventional procedures for placing smart components such as sensors, actuators, processing equipment, etc., can be cumbersome or unnecessarily invasive.

SUMMARY

[0003] The needs above, as well as others, are addressed by embodiments of apparatuses for providing feedback on implants, as well as systems for providing implant feedback, and related methods described in this disclosure. All examples and features mentioned below can be combined in any technically possible way.

[0004] Various implementations include orthopedic implants and related electrode assemblies adapted for use with such orthopedic implants. In a particular implementation, an electrode assembly includes: a first housing containing at least one electrode; and a control module with a first connector for mating with a complementary connector on the first housing, the control module including a controller programmed to initiate transmission of at least one of a measurement pattern or a stimulation pattern to a patient with the at least one electrode, where the first housing and the control module include a biocompatible material, and when connected, are sized to complement an orthopedic implant.

[0005] In particular aspects, an orthopedic implant includes: a body for securing in a patient, the body including a mating feature; and an electrode assembly coupled with the body at the mating feature, the electrode assembly including: a first housing containing at least one electrode; and a control module connected with the first housing, the control module including a controller programmed to initiate transmission of at least one of a measurement pattern or a stimulation pattern to a patient with the at least one electrode, where the electrode assembly complements the mating feature on the body.

[0006] Implementations may include one of the following features, or any combination thereof.

[0007] In some examples, the biocompatible material includes an electrically insulative biocompatible material. In particular examples, the biocompatible material includes a rigid material such as PEEK (polyether ether ketone), HXLPE (highly cross-linked polyethylene) and/or a flexible material such as silicone, TPU (thermoplastic polyurethane), or PTFE (polytetrafluoroethylene). In additional cases, the biocompatible material includes an implantable metal such as stainless steel or titanium, that is electrically isolated from the electrodes.

[0008] In certain examples, the controller is programmed to initiate transmission of both the measurement pattern and the stimulation pattern to the patient, and the at least one electrode is configured for use in both the measurement pattern and the stimulation pattern.

[0009] In particular cases, the first housing contains a plurality of electrodes.

[0010] In some aspects, the electrode assembly further includes a second housing containing at least one additional electrode, where the control module includes a second connector for mating with a complementary connector on the second housing.

[0011] In certain implementations, when the first housing and the control module are connected, the electrode assembly is pliable to complement at least one contour in the orthopedic implant.
[0012] In some cases, the electrode assembly is sized to sit within a groove or slot in the orthopedic implant.
[0013] In particular aspects, the first housing includes a flexible insulative section adjacent the at least one electrode.
[0014] In some implementations, the first housing includes a plurality of electrodes interposed between adjacent insulative sections, where each of the plurality of electrodes is selectable for at least one of: measurement based on the measurement pattern or stimulation based on the stimulation pattern.
[0015] In certain cases, the plurality of electrodes includes at least three electrodes, where according to the stimulation pattern: a first electrode is selected as an anode and a second electrode is selected as a cathode. In certain cases, additional electrodes are selected as one of an anode or a cathode according to the stimulation pattern. In accordance with the measurement pattern: a first electrode is selected as a working electrode, a second electrode is selected as a counter electrode, and a third electrode is selected as a reference electrode. In some examples, the first, second, and third electrode form an electrochemical cell in the measurement pattern.
[0016] In particular cases, the controller is configured to change selection of one or more of the plurality of electrodes according to the stimulation pattern or the measurement pattern.
[0017] In certain aspects, the first housing includes a flexible tubular body and the at least one electrode is integrated in a wall of the flexible tubular body such that a primary axis of each electrode is parallel with a primary axis of the flexible tubular body. In various examples, the primary axis is the long axis.
[0018] In some cases, the first housing includes a flexible tubular body with an internal channel, and the at least one electrode is integrated in a wall of the flexible tubular body such that a primary axis of each electrode is off-axis relative to a primary axis of the flexible tubular body. In some cases, the internal channel contains wiring, e.g., internal wiring. In other cases, wiring is integrated in or otherwise connected with the walls. In certain cases, multiple electrodes are integrated in a wall of the flexible tubular body and are either evenly spaced relative to one another or are unevenly spaced relative to one another. In various examples, the off-axis electrodes are approximately parallel to the primary axis of the flexible tubular body.
[0019] In certain cases, the control module includes one or more application specific integrated circuits (ASICs).
[0020] In particular cases, the control module includes a hermetically sealed housing constructed from an implantable metal or a biocompatible polymer. In some cases, the implantable metal includes titanium. In certain examples, the biocompatible polymer includes PEEK. In various implementations, the housing includes: the controller, memory, a multiplexer, a power system, and a communication system, wherein the power system enables wireless power transfer to the control module and the communication system enables wireless communication with an external control device. In some cases, wireless communication includes communication with one or more protocols, for example, Bluetooth (BT), BT Low Energy (BLE), radio frequency (RF), etc. In particular cases, the control module is connected with a local control and/or display device such as a smart phone or smart device (e.g., tablet, computing device, etc.) that has network connectivity. In some examples, the local control and/or display device can be configured to communicate over a network to transmit treatment instructions to the control module and/or receive sensor data for remote transmission.
[0021] In certain aspects, the first housing and the control module have a substantially arcuate outer dimension or a substantially angled outer dimension. In some cases, the substantially angled outer dimension includes a tubular outer dimension, a rectangular outer dimension, or an oblong outer dimension. In particular cases, each electrode can have a shape corresponding with an outer

dimension of the control module. In particular cases, the combined assembly has a linear shape or a toroidal shape. In certain cases, the shape of electrodes and/or the control module can be tailored to fit in the orthopedic implant, e.g., within a slot or opening in the implant. For example, a slot with angled walls or arcuate walls can be fit with an assembly that has angled outer walls or arcuate outer walls.

[0022] In some examples, the orthopedic implant includes a knee implant, a tibial insert, a hip implant, or a surgical nail.

[0023] In particular cases, the first housing and the control module are integrally assembled, or configured for assembly during a surgical procedure on the patient.

[0024] In other examples, the electrode assembly and control module are provided separately and assembled during the surgical procedure, e.g., in the surgical theater or intraoperatively. In certain examples, each electrode assembly has an electrical connector component with one or more conductive surfaces (e.g., pads) that are individually electrically coupled to each electrode. In certain cases, the electrode assembly is securely attached to mating contacts on the control module housing (or, canister), via mechanical means such as with set screws or pins.

[0025] In certain aspects of the orthopedic implant, the first housing includes a plurality of electrodes interposed between adjacent insulative sections, where each of the plurality of electrodes is selectable for at least one of: measurement based on the measurement pattern or stimulation based on the stimulation pattern.

[0026] In further aspects of the orthopedic implant, the controller is programmed to initiate transmission of both the measurement pattern and the stimulation pattern to a patient, and the at least one electrode is configured for use in both the measurement pattern and the stimulation pattern.

[0027] In additional aspects of the orthopedic implant, the mating feature includes a groove or a slot in the body, and the electrode assembly is retained in the groove or slot in the body.

[0028] Two or more features described in this disclosure, including those described in this summary section, may be combined to form implementations not specifically described herein.

[0029] The above presents a simplified summary in order to provide a basic understanding of some aspects of the claimed subject matter. This summary is not an extensive overview. It is not intended to identify key or critical elements or to delineate the scope of the claimed subject matter. Its sole purpose is to present some concepts in a simplified form as a prelude to the more detailed description that is presented later.

[0030] The details of one or more implementations are set forth in the accompanying drawings and the description below. Other features, objects and benefits will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0031] FIG. **1** shows a schematic system view of an implant according to various implementations.

[0032] FIG. **2** shows an example electrode assembly according to various implementations.

[0033] FIG. **3** shows an example electrode assembly according to various additional implementations.

[0034] FIGS. **4**A-**4**C show example electrode assemblies according to various further implementations.

[0035] FIG. **5** shows a perspective view of circular or oblong electrode assembly according to various implementations.

[0036] FIG. **6** shows a perspective view of an electrode assembly including two distinct electrode housings according to various implementations.

[0037] FIG. **7** is a system diagram of a control module according to various implementations.
[0038] FIG. **8** illustrates a variation on the control module of FIG. **7** according to various implementations.
[0039] FIG. **9** shows a tubular form of an electrode housing according to various implementations.
[0040] FIG. **10** shows a rectangular form of an electrode housing according to various additional implementations.
[0041] FIGS. **11**A-**11**C and **12**A-**12**C illustrate control functions in electrode measurement and/or stimulation according to various implementations.
[0042] FIGS. **13**A-**13**D show applications of electrode assemblies in an implant according to various implementations.
[0043] FIG. **14** is a perspective view of an electrode assembly for an implant according to various additional implementations.
[0044] FIGS. **15**A-**15**B show views of an implant with a slot or recess for receiving an electrode assembly according to various implementations.
[0045] FIG. **16** shows an electrode assembly in an implant according to one of various implementations.
[0046] FIG. **17** shows an electrode assembly in an implant according to various additional implementations.
[0047] FIGS. **18**A-**18**B show an electrode assembly in two distinct implants according to various additional implementations.
[0048] FIGS. **19**A-**19**B illustrates flexible tubular electrode assemblies in two distinct implants according to various additional implementations.
[0049] FIG. **20** shows a type of flexible tubular electrode assembly according to various implementations.
[0050] FIG. **21** shows another type of flexible tubular electrode assembly according to various implementations.
[0051] FIG. **22** shows a system for enabling control of an implant according to various implementations.
[0052] It is noted that the drawings of the various implementations are not necessarily to scale. The drawings are intended to depict only typical aspects of the disclosure, and therefore should not be considered as limiting the scope of the implementations. In the drawings, like numbering represents like elements between the drawings.
DETAILED DESCRIPTION
[0053] Various example embodiments of assemblies and implants with integrated electrodes and control modules for controlling transmission of a measurement pattern and/or a stimulation pattern to a patient are described herein. Various particular implementations include orthopedic implants such as a knee implant, a tibial insert, a hip implant, or a surgical nail with an integrated electrode assembly.
[0054] In the interest of clarity, not all features of an actual implementation are necessarily described in this specification. It will of course be appreciated that in the development of any such actual embodiment, numerous implementation-specific decisions must be made to achieve the developers' specific goals, such as compliance with system-related and business-related constraints, which will vary from one implementation to another. Moreover, it will be appreciated that such a development effort might be complex and time-consuming, but would nevertheless be a routine undertaking for those of ordinary skill in the art having the benefit of this disclosure. The apparatuses and related systems and methods described herein boast a variety of inventive features and components that warrant patent protection, both individually and in combination.
[0055] It is to be understood that any given elements of the disclosed embodiments of the invention may be embodied in a single structure, a single step, a single substance, or the like. Similarly, a given element of the disclosed embodiment may be embodied in multiple structures, steps,

substances, or the like.

[0056] As noted herein, the various disclosed implementations can reduce complexity and/or invasiveness of conventional surgical procedures, for example, by attaching the electrodes on or within the surface of the implant form. Further, relative to conventional implants, the disclosed implementations efficiently allocate electrodes and control modules, and also provide one or more separate components that can be assembled to traditional implant components within the surgical theater. Various implementations include assemblies, implants, systems and methods of coupling electrodes to smart orthopedic implant systems.

[0057] This disclosure provides, at least in part, an electrode assembly sized to complement an orthopedic implant and a related orthopedic implant. Various implementations of the electrode assembly allow a medical professional such as a surgeon to insert or otherwise affix the electrode assembly to an orthopedic implant, e.g., in a surgical theater or intraoperatively. Various implementations include an implant with an electrode assembly that can be used to initiate at least one of a measurement pattern or a stimulation pattern to a patient. The various disclosed implementations can improve patient outcomes when compared with conventional implants. The disclosed implementations can provide post-operative feedback on implant procedures and healing, as well as a means to stimulate healing post-operatively. The disclosed implementations can enhance both current procedural outcomes as well as future surgical outcomes.

[0058] Commonly labeled components in the FIGURES are considered to be substantially equivalent components for the purposes of illustration, and redundant discussion of those components is omitted for clarity.

[0059] FIG. **1** shows a schematic depiction of an orthopedic implant **10** according to various implementations. Variations on the implant **10** are illustrated in FIGS. **2**-**6** as implants **10**A-**10**G. The implants **10** are referred to collectively, and can include an electrode assembly **15** that has a first housing **20** including at least one electrode **30**, and a control module **40** with a first connector **50** for mating with a complementary connector **60** on the first housing **20**. In various implementations, the first housing **20** and the control module **40** include a biocompatible material. In some examples, the biocompatible material includes an electrically insulative biocompatible material. In particular examples, the biocompatible material includes a rigid material such as PEEK (polyether ether ketone), HXLPE (highly cross-linked polyethylene) and/or a flexible material such as silicone, TPU (thermoplastic polyurethane), or PTFE (polytetrafluoroethylene). In additional cases, the biocompatible material includes an implantable metal such as stainless steel or titanium, that is electrically isolated from the electrode(s) **30**. As described further herein, the control module **40** can include a controller that is programmed to initiate transmission of a measurement pattern and/or a stimulation pattern to a patient with the electrode(s) **30**. In various implementations, the control module **40** includes a hermetically sealed housing constructed from an implantable metal or a biocompatible polymer. In some cases, the implantable metal includes titanium. In certain examples, the biocompatible polymer includes PEEK.

[0060] In certain cases, the first housing **20** includes a plurality of electrodes **30**. In additional implementations, the implant includes a second housing **70** containing at least one electrode **30**, and the control module **40** includes a second connector **80** for mating with a complementary connector **90** on the second housing **70**. Connectors can include electrical connectors for enabling electrical connection between electrodes **30** and the control module **40**, as well as physical connectors such as clasps, screws, pins, snap-to-fit connectors, press-to-fit connectors, etc. In certain cases, electrical connectors are integrated into physical connectors. FIG. **1** shows an example implant **10** including a control module **40** that is coupled with two distinct housings **20**, **70**, each including at least one electrode **30** for executing a measurement pattern and/or stimulation pattern, which may including transmitting the pattern to a patient. FIG. **2** shows an example implant **10**A with a control module **40** coupled with a first housing **20** that includes at least one electrode (not shown) for executing a measurement pattern and/or stimulation pattern to a patient.

Implant **10**A can include electrode(s) **30** in a housing **20** that has an arcuate outer dimension **90**, e.g., an arcuate outer surface such as a tubular shape having a substantially round or substantially oblong cross sectional geometry. Similarly, the control module **40** can have an arcuate outer dimension **90**, e.g., a tubular shape. In certain cases, the control module **40** has a greater outer dimension **90** (or, outer diameter) than that of the electrode housing **20**.

[0061] FIG. **3** shows an implant **10**B with a control module **40** coupled with two distinct housings **20**, **70** on a common side **110** of the control module **40**, also for executing a measurement pattern and/or stimulation pattern to a patient. Implant **10**B can include electrode housings **20**, **70** that have an arcuate outer dimension **90** and a control module **40** having a polyhedral or substantially polyhedral geometry with an outer dimension **120**, e.g., an approximately cube, rectangular prism, or three dimensional rounded rectangular prism shape. In certain of these cases, the outer dimension **120** of the control module **40** is greater than the outer dimension **90** (e.g., outer diameter) of each housing **20**, **70**, and in further cases, is greater than the collective outer dimension of both housings **20**, **70**. FIGS. **4**A-**4**C show distinct configurations of implants **10**C, **10**D, and **10**E, respectively, with both housing(s) **20**, **70** and control modules **40** with angled outer dimensions **130**, e.g., squared or rectangular outer dimensions. In certain of these cases, at least one dimension of the housing **20**, **70** and control module **40** is greater than another dimension, e.g., a length is greater than a width and/or a depth, or a length is greater than both a width and a depth, or a length is greater than a width, which are both greater than a depth. In certain of these cases, the housing **20**, **70** is connected to the control module **40** in a linear arrangement such as, e.g., FIGS. **4**A and **4**B. In particular examples, such as FIG. **4**B, the control module **40** is interposed between electrode assembly housings **20**, **70**. In other examples, such as FIG. **4**C, the control module **40** is connected to the electrode assembly housing **20** at two locations **140**, **150**, forming a ring shape. FIG. **5** shows another variation on an implant **10**F in a ring shape with a central control module **40** and an electrode assembly housing **20** that connects to the housing **20** at two locations **140**, **150**. In the example implant **10**F, the outer dimension **160** of housing **20** and control module **40** is arcuate, e.g., rounded or otherwise non-angular. FIG. **6** shows implant **10**G that is similar in linear arrangement to implant **10**D (FIG. **4**B), but has an arcuate outer dimension **170** in both the housings **20**, **70** and control module **40**, e.g., the outer dimension **170** is rounded or otherwise non-angular. Various additional form factors of implant **10** are illustrated herein. It is understood that distinct form factors can be combined to provide an implant according to the various implementations herein. Further, while electrode assembly housing(s) **20**, **70** are described herein, it is understood that a one or a plurality of electrodes can be independently selected for containment in each housing. For example, a first housing **20** can contain a plurality of electrodes for performing functions described herein. In various implementations, the housing(s) **20**, **70** include connectors such as conductive connectors for coupling with the control module **40** and enabling control functions described herein.

[0062] In any case, once connected, the first housing **20** (and in some cases, second housing **70** and/or additional housings) and control module **40** of the electrode assembly **15** are sized to complement an orthopedic implant **10**. That is, the orthopedic implant **10** can integrate, seat, adhere to, or otherwise retain the electrode assembly **15** to enable implanting and control of the application and/or transmission of measurement and/or stimulation pattern to the patient.

[0063] FIG. **7** shows a schematic illustration of example electronics in a control module **40** according to various implementations. In various implementations, the control module **40** includes a controller **200**, a power system **210**, and a communication system **220**. In some examples, the controller **200** includes one or more microcontrollers or microcontroller units (MCU), which in certain cases can include a programmable processor configured (e.g., programmed) to perform functions described herein. In some examples, the controller **200** can include one or more integrated circuits that include one or more of a processor, memory module, communication interfaces, and programmable inputs and outputs. In certain cases, the power system **210** includes a power distribution subsystem **230**, power storage **240** (e.g., a battery or capacitor device), and a wireless power transfer (WPT) system **250** including, e.g., a WPT receiver/transmitter **260** such as a WPT antenna or coil, and a WPT subsystem **270**. In various implementations, the power system **210** enables WPT to the control module **40** from an external device such as an external power source, e.g., a charging device, a radio frequency (RF) device, etc. The communication system **220** can include a communication antenna **280** and a communication subsystem **290** for managing communication signals from the antenna **280**. The communication system **220** enables wireless communication with an external control device, e.g., a smart device such as a smart phone, tablet, computing device, etc. In some cases, wireless communication includes communication with one or more protocols, for example, Bluetooth (BT), BT Low Energy (BLE), radio frequency (RF), etc. The control module **40** can also include an electrode control system **300** that includes a multiplexer **310** for interfacing with electrode interface connectors **320**, and in certain cases, a potentiostat controller **330** for controlling execution of the multi-electrode stimulation and/or measurement patterns from the controller **200**. As shown, the control module **40** can also include memory **390** (e.g. electrically erasable programmable read-only memory (EEPROM)) for storing instructions such as orthopedic implant measurement and/or stimulation patterns, e.g., instructions for executing a measurement pattern and/or a stimulation pattern via the potentiostat controller **330** and the multiplexer **310**. In various implementations, the controller **200** is programmed to initiate transmission of a measurement pattern and/or a stimulation pattern to a patient with at least one of the electrodes **30** (FIGS. **1-6**) via the electrode interface connectors **320**.

[0064] In certain examples, the multiplexer **310** enables individual electrodes to be configured with one or more roles such as the anode or cathode for electrical stimulation functions, or as a working electrode, reference electrode, or counter electrode for electrochemical sensing functions. The multiplexer **310** interfaces with the potentiostat controller **330** which handles the digital-to-analogy conversion of stimulation signals and analog-to-digital conversion of sensor response signals. The potentiostat controller **330** enables the control module **40** to perform a variety of electrochemical measurements using voltammetric techniques such as linear sweep voltammetry, cyclic voltammetry, square wave voltammetry, differential pulse voltammetry, and/or normal pulse voltammetry as well as time-based techniques such as chronoamperometry, pulsed amperometry, open circuit potentiometry, and/or electrochemical impedance spectroscopy. The potentiostat controller **330** can also control the stimulation output in various profiles such as current-controlled, voltage-controlled, monophasic pulsed direct current (DC), biphasic pulsed DC, and alternating current (AC).

[0065] In particular cases, the control module **40** is connected with a local control and/or display device such as a smart phone or smart device (e.g., tablet, computing device, etc.) that has network connectivity. In some examples, the local control and/or display device can be configured to communicate over a network to transmit treatment instructions to the control module **40** and/or receive sensor data for remote transmission, e.g., via a network and/or via a cloud-based data connection.

[0066] According to various implementations, the controller **200** is configured (e.g., programmed) to initiate transmission of one or both of a measurement pattern or a stimulation pattern to a patient. In certain cases, the controller **200** is configured (e.g., programmed) to initiate transmission of both a measurement pattern and a stimulation pattern to a patient. As described further herein, a given electrode **30** can be configured for use in both the measurement pattern and the stimulation pattern.

[0067] FIG. **8** shows a variation on the control module **40** whereby certain components are integrated in one or more application specific integrated circuits (ASICs) **400**. For example, an ASIC can house one or more components in the controller **200**, power system **210**, and/or communication system **220**. In certain cases, features such as the multiplexer **310**, potentiostat controller **330**, and/or power regulator **420** are housed on the ASIC **400**. Further, an interface **430** for a transmitter/receiver antenna/coil **440** can be housed on the ASIC **400** in certain implementations. The use of ASIC(s) **400** can enable miniaturization of certain components in the control module **40**, which can enhance adaptability in terms of fit and/or coupling with the body of the implant **10**.

[0068] FIG. **9** shows an example schematic depiction of a housing **20** including at least one electrode **30** according to various implementations. In this implementation, the housing **20** includes a body **500** with one or more integrated conductive elements (i.e., electrodes **30**) that are electrically isolated from one another. In certain cases, the body **500** includes a channel **510** such as an internal channel that extends along the primary axis (A.sub.pB) of the body **500**. In certain cases, the channel **510** houses one or more wires and/or conductive traces that enable connection of each electrode **30** with the control module **40**. In certain cases, the body **500** includes a single continuous element with terminations at each end, or in other cases, is a toroidal shape (e.g., a ring or loop). In various implementations, multiple electrodes **30** are positioned along the body **500**, e.g., two, three, four, five, or more electrodes **30**. In certain cases, the body **500** includes an electrically insulative biocompatible material that is rigid (e.g. PEEK, HXLPE, etc.) or flexible (e.g. silicone, TPU, PTFE, etc.). Alternatively, the body **500** may be constructed from an implantable metal (e.g. stainless steel, titanium, etc.) that is electrically isolated from the electrodes **30**. The electrodes **30** may be constructed from a noncorrosive conductive implantable metal (e.g. titanium, nitinol, platinum-iridium alloy) or biocompatible carbon-based material (e.g. carbon, graphite, graphene). The exposed surface of the electrode **30** may also include a structural base material (e.g. stainless steel, titanium) coated with a noble metal (e.g. gold, platinum, silver) or carbon-based material (e.g. carbon, graphite, graphene). Additionally, one or more of the electrodes **30** may be functionalized by adding an enzyme layer to couple antibodies, DNA, or proteins that can bind with the target analyte (e.g. bacterial cell) and produce an electrochemical response that can be sensed. In certain cases, as discussed herein, the body **500** can be pliable or flexible to complement distinct shapes and/or slots/grooves in an implant **10**. In various implementations, the body **500** is insulative, and includes distinct insulative sections **500**A, **500**B, **500**C, etc., that are interposed between adjacent electrodes **30**A, **30**B, **30**C, etc. In certain cases, electrodes **30** protrude from or extend beyond an outer dimension of the body **500** in at least one direction. For example, as shown in FIG. **9**, the outer surface **530** of the electrode **30** extends radially beyond an outer surface **540** of the body **500** (relative to primary axis (A.sub.pB)). In certain cases, the body **500** is compatible with arcuate embodiments of the electrode assembly, e.g., as shown in FIGS. **2**, **3**, **5**, and **6**.

[0069] FIG. **10** shows another implementation of a housing **20** including electrodes **30** that are retained in a body **600** that has one or more angular outer surfaces **630**, e.g., one or more edges or sides. In certain cases, a plurality of electrodes **30** are arranged linearly along the body **600**. The body **600** includes channel **610**, such as an internal channel extending along the primary axis (A.sub.pB) of the body **600**. In certain cases, the channel **610** houses one or more wires and/or conductive traces that enable connection of each electrode **30** with the control module **40**. In various implementations, the body **600** is formed of a flexible or pliable material that enables the housing **20** to be bent, twisted, or otherwise manipulated to fit a slot/groove/opening in an implant **10** and/or to comply with a curvature in a surface of an implant **10**. In the example implementation of body **600**, the electrodes **30** may be evenly or unevenly spaced along the length of the body **600**. In certain cases, electrodes **30** coupled with body **600** are rectangular or square electrodes that are evenly spaced along the length of the body **600** (e.g., along primary axis (A.sub.pB)). In other cases, the electrodes **30** may be rectangular, circular, or polygonal shaped. In various implementations, the body **600** is insulative, and includes distinct insulative sections **600**A, **600**B, **600**C, etc., that are interposed between adjacent electrodes **30**A, **30**B, **30**C. In other cases, the body **600** is formed of a single component with openings for adjacent electrodes **30**A, **30**B, **30**C. In particular cases, outer surfaces **630** of the body **600** are approximately flush with outer surfaces **640** of the electrodes **30**, or the outer surfaces **640** of electrodes **30** are recessed relative to the outer surfaces **630** of the body **600**. In these examples, the outer surfaces **640** of the electrodes **30** do not extend beyond the outer surfaces **630** of the body **600**. In certain cases, the body **600** is compatible with angular embodiments of the electrode assembly, e.g., as shown in FIGS. **4**A-**4**C.

[0070] In various implementations, the electrodes **30** (e.g., FIGS. **1-6**, **9**, **10**) are selectable for measurement based on a measurement pattern and/or stimulation based on a stimulation pattern. FIGS. **11-11**C illustrate an implementation of a housing **20** including a plurality of electrodes **30** (e.g., three electrodes) separated (or, insulated) from one another by sections **500**A of the body **500**, with distinct electrode assignments based on a stimulation pattern. With reference to FIG. **11**A, a first assignment includes assigning positive polarity to first and third electrodes **30**A, **30**C and negative polarity to second electrode **30**B that is positioned between the first and third electrodes **30**A, **30**C. A second assignment (FIG. **11**B) includes assigning negative polarity to first and third electrodes **30**A, **30**C and positive polarity to second electrode **30**B. A third assignment (FIG. **11**C) includes assigning positive polarity to first electrode **30**A and negative polarity to second and third electrodes **30**B, **30**C. In various implementations, positive electrodes act as anodes and negative electrodes act as cathodes. It is understood that during use, the electrode **30** transmits signals from the anode(s), through a material of interest (or, material under test), which can be received (or, detected) at the cathode(s).

[0071] FIGS. **12**A-**12**C show another implementation of a housing **20** including a plurality of electrodes **30** (e.g., five electrodes) separated (or, insulated) from one another by sections **500**A-E of the body **500**, with distinct electrode assignments based on an applied measurement pattern. It is understood that this housing **20** can be used as part of an electrochemical cell for conducting measurement on a patient. Further, it is understood that this housing **20** can be used for both stimulation as described with respect to FIGS. **11**A-**11**C, as well as for measurement, e.g., with the controller **200** dictating the polarity and/or assignment of electrodes **30** for use in one or more patterns.

[0072] In the example of FIGS. **12**A-**12**C, each electrode **30** can be directed as the working electrode (WE), a counter electrode (CE), or reference electrode (RE) to form an electrochemical cell and enable sensor measurements at each electrode location in the assembly. An example sequential chain of changing electrode role is presented in the progression from FIG. **12**A, to FIG. **12**B, to FIG. **12**C, whereby the role of electrodes **30**A-E change according to a prescribed measurement pattern. In this example, in progression from FIG. **12**A to FIG. **12**C, the working electrode is switched from electrode **30**B to electrode **30**C, and then to electrode **30**D, while counter electrodes and reference electrodes are switched from **30**A to **30**B to **30**C and **30**C to **30**D to **30**E, respectively. In various implementations, when performing measurement (e.g., electrochemical sensing) functions, each electrode **30** of the assembly may be used as the working electrode (i.e. sensing electrode), while the adjacent electrodes act as the counter electrode (CE) and reference (or pseudo-reference) electrode (RE) to complete the electrochemical cell. In particular examples, measurements can be captured in series at every electrode **30** by alternating which electrodes **30**A, **30**B, **30**C, etc. are used as the working, counter, and reference electrodes. In a similar manner, as described with reference to housing **20** in FIGS. **9-11**, each electrode **30** may serve as the anode or cathode in a two-electrode stimulation treatment system, e.g., to apply a stimulation pattern. By adjusting the stimulation program and directing which electrodes **30** are the anode and cathode, the user/operator can design a stimulation protocol to target a treatment based on clinical need.

[0073] FIGS. **13**A-**13**D illustrate various depictions of an implant **10** including integrated electrode assemblies **15** according to various implementations. In this implementation, the implant **10** includes a knee implant such as a total knee implant. It is understood that the electrode assemblies **15** illustrated and described according to various implementations can also be applied to various other implants. For example, in addition to knee implants, the assemblies **15** can be particularly suited for integration with at least one of a tibial insert, a hip implant, or a surgical nail.

[0074] In these examples, implant **10** can have at least four distinct integration features, shown in FIGS. **13**A through **13**D for mounting, adhering, or otherwise affixing the electrode assemblies **15** illustrated herein. Various other integration features are also possible. In the examples shown, one or more cylindrical electrode assembles **15** may be arranged in or on the surface **700** of an orthopedic implant **10**. For example, the cylindrical assembly **15** can fit within a groove or slot **710** in the implant body **720**. In certain cases, the groove or slot **710** includes a semicircular groove or slot, and/or includes a lip for retaining the assembly **15**. In particular examples, the assembly **15** is sized to enable a snap-to-fit or press-to-fit connection with the body **720** at the groove or slot **710**. In certain examples, the groove or slot **710** is recessed from the outermost surface of the body **720**, such that a portion of the assembly **15** (e.g., an electrode) does not protrude from the outermost surface of the body **720**. In particular examples, the assembly **15** sits approximately flush with the outermost surface of the body **720** to enable contact with the patient's tissue once implanted. In some cases, each assembly **15** may be preassembled with the implant components as part of the manufacturing assembly process. Alternatively, each electrode assembly **15** may be manufactured apart from the implant components and assembled by the surgical team in the operating theater. In particular cases, the assembly **15** includes at least one flexible insulative section (e.g., such as sections **500**A, **500**B, etc., FIG. **9**) that enables the assembly **15** to complement contours in the implant body **720**. In certain cases, the flexible insulative sections enable an operator (e.g., medical professional) to adjust the contour of the assembly **15** to the implant body **720** during or prior to a surgical procedure. In further examples, components of an electrode assembly **15** (including housing **20** and control module **40**) are provided separately and assembled during the surgical procedure, e.g., in the surgical theater or intraoperatively. In certain examples, as noted herein, each housing **20** has an electrical connector component with one or more conductive surfaces (e.g., pads) that are individually electrically coupled to each electrode **30**. In certain cases, the housing **20** is securely attached to mating contacts on the control module **40**, e.g., on the housing or canister of the control module **40** via mechanical means such as with set screws or pins, connections of which can be made in the surgical theater or intraoperatively.
[0075] FIG. **14** shows a detailed view of an electrode assembly **15** such as is shown in FIG. **13**D, including a flexible housing **20** (including body sections **500** and electrodes **30**) coupled with a control module **40** at two sides to form a flexible band. A flexible band implementation can be beneficial because it is compliant to match the curvature of the tibial insert geometry, can snap into a mating feature (e.g. groove or slot) on the tibial insert, and can stretch to fit a variety of tibial insert sizes and types. In various implementations, the flexible band enables both adjustment of shape and size, e.g., it can be shaped to fit the geometry of an implant and also expand (or flex) to fit around portions of an implant. In certain cases, the housing **20** has elastic properties enabling a stretch-to-fit application of the assembly **15**. In various implementations, the flexible band assembly, e.g., assembly **15** in FIG. **14** can attach to a tibial insert via a mating feature machined into the outer surface of the implant body. In certain cases, the mating feature is machined to complement a size of the assembly **15**, e.g., an outer dimension of the assembly **15**.
[0076] A particular depiction of a mating feature **800** in an implant **10** is illustrated in FIGS. **15**A and **15**B, which show a tibial implant in side and cross-sectional views, respectively. The mating feature **800** in this example includes a semi-circular groove or slot **810** in the body **820**. A semi-circular groove or slot can be beneficial for creating a cavity to house and retain the assembly **15**. A retention feature **830** such as a protruding edge on the upper portion **840** and/or lower portion **850** of the groove or slot **810** enables retention of the assembly **15**, e.g., in a snap-to-fit or press-to-fit application. During assembly, the assembly **15** can be deformed (without damage) to fit between the retention feature **830** to snap into the groove or slot **810**. The retention features **830** prevent unwanted dislodging of the assembly **15** during use. In some cases, removal of the assembly **15** from the groove or slot **810** can only be performed with a tool. FIG. **16** illustrates a flexible band-type assembly **15** (FIG. **14**) assembled with the tibial insert implant **10** (FIGS. **15**A-**15**B) according

to various implementations. FIG. **17** shows the flexible band-type assembly **15** (FIG. **15**) assembled with the knee implant of FIG. **13**C.

[0077] Certain applications, such as on the rim of the femoral component shown in FIG. **18**A, may benefit from a cylindrical electrode body **30** with an integrated control module **40** similar to configurations of an electrode assembly **15** shown in FIGS. **2** and **3**. Other applications of such an assembly **15** configuration include hip arthroplasty and trauma nailing. In a total hip replacement application such as illustrated in the hip implant **900** shown in FIG. **18**A, the mating feature (e.g., slot or groove) **910** can be located along the length of the femoral stem **920** to accept an electrode assembly **15** such as a cylindrical electrode body, or the mating feature **910** can be located along circumferential edge **930** of the acetabular cup **940** to accept a flexible band electrode assembly **15**. In a trauma nailing application, such as a troch nail **1000** shown in FIG. **18**B, a mating feature **1010** can be located along the length of the nail shaft **1020** to accept a cylindrical electrode assembly **15** with integrated control module **40**.

[0078] In any of the previously described applications and arrangements, it may be preferable for the form of the electrode assembly **15** (including integrated control module **40**) to be of a non-circular cross-section, e.g., as illustrated in FIG. **4**. As noted herein, one alternate embodiment may be a rectangular cross-section.

[0079] In certain other applications, it may be preferable for one or more of the previously described electrode assemblies to be integrated in a flexible tubular housing. A tubular embodiment can enable a flexible tubular assembly to be attached to a shaft-like implant component without the need for mating features to be machined for each electrode assembly. In a total hip replacement application, e.g., as shown in FIG. **19**A, tubular electrode assemblies **1100** may be attached around one or more of the neck **950**, superior portion **960**, or inferior portion **970** of the femoral stem **920**. In a trauma nailing application (FIG. **19**B), such as a troch nail **1000**, the tubular electrode assemblies **1100** may attach around the superior **1040** or inferior **1050** portions of the nail shaft **1020**. In both applications, a feature on the implant body may be beneficial to prevent axial migration of tubular electrode assembly **1100** with respect to the implant component during and after implantation. One such feature may consist of a recess or trench **1060** in which the flexible tubular electrode assembly **1100** can sit. In certain cases, the recess or trench **1060** is wider than a width of the electrode assembly **1100**, which can enable the assembly **1100** to be wrapped around the body and retained by its own elasticity within the recess or trench **1060**. In certain examples, the recess or trench **1060** is defined by at least two edges or sides that can axially retain the assembly **1100**.

[0080] An example of a first type of flexible tubular assembly **1100**A is illustrated in FIG. **20**, including a flexible body **1110**, and a plurality of cylindrical electrode assemblies **15** similar to electrode assemblies described herein. The electrode assemblies **15** can each include a set of body members **500** interposed between electrodes **30**, and in certain cases, each coupled with a control module **40**. In other cases, a central or shared control module **40** can be positioned within the body **1110** (e.g., in a channel **1120**) or at a terminal end of the body **1110** and coupled with one or more electrode assemblies **15**. In certain cases, a plurality of cylindrical electrode assemblies **15** are coupled with a wall **1130** (or walls) of the body **1110**. In certain examples, at least one electrode **30** is integrated in the wall **1130** of the flexible tubular body **1110** such that a primary axis of each electrode is approximately parallel with a primary axis (A.sub.ftb) of the flexible tubular body **1110**.

[0081] In another implementation, as shown in FIG. **21**, a flexible tubular assembly **1100**B includes a flexible body **1210** and a plurality of annular electrodes **1220** arranged around the flexible body **1210**. In these implementations, the body **1210** includes an internal channel **1230**, and at least one of the electrodes **1220** is integrated in a wall **1240** of the body **1210** such that a primary axis of the electrode **1220** is off-axis relative to a primary axis (A.sub.ftb) of the body **1210**, e.g., approximately perpendicular to at least 5 degrees off-axis. The example in FIG. **21** shows electrodes **1220** running perpendicular to the primary axis (A.sub.ftb) of the body **1210**. In certain cases, adjacent electrodes **1220** are evenly separated (spaced) relative to one another, and in other cases, are unevenly separated (spaced). In particular examples, the electrodes **1220** are integrated directly within the walls **1240** of a flexible tubular body **1210**, e.g., in slots or gaps. In addition, the channel **1230** can be configured to house (and in some cases, insulate) wires or contacts between each electrode and an integrated control module. In certain of these cases, a single control module (not shown) is housed in the channel **1230** to control the electrodes **1220**. In a particular example, wire channels **1250** are integrated in the walls **1240** (e.g., extending axially along (A.sub.ftb) enabling routing of wires **1260** between electrodes **1220** and the control module(s).

[0082] In particular cases, the implants **10** shown and described herein can be part of a "smart" implant system or platform, as illustrated in the system **1300** in FIG. **22**. In these implementations, a patient **1310** having an implant **10** such as those described herein can be provided with a diagnostic or interventional treatment such as a measurement and/or sensing treatment via a smart device **1320**, which may be connected via a network such as a cloud server **1330** with a clinician dashboard **1340**. The dashboard **1340** can include an interface enabling a clinician to send a treatment activation signal through the network **1330** to the smart device **1320** to activate treatment at the implant **10** in the patient **1310**. The dashboard **1340** can also be configured to receive data such as sensor data, charge status, and/or patient reported data (via network **1330**) from the smart device **1320**, which in turn receives sensor data and/or charge status data from the implant **10**. In one implementation of a smart implant control system, the implant **10** communicates with the patient's smart device **1320** over a 2.4 GHz protocol such as Bluetooth or Bluetooth Low Energy. A smart device-compatible dongle may also be used to communicate to the implant **10** on a radiofrequency (RF) band that is not directly supported by the smart device **1320**, such as 915 MHz or 401-406 MHz (MICS). The smart device dongle can contain transmission and receiver antenna(s) tuned to the target frequency of the implant communication protocol. In various examples, data is sent from patient's implant **10**, received by the integrated antenna or dongle's antenna at the smart device **1320**, and then processed through a dedicated software application (or, app) on the patient's smart device **1320**. The data can then be uploaded from the patient's smart device **1320** to a secure server/network via internet connection such as WiFi or LTE. Data may be processed on the patient's smart device **1320** prior to upload or on the server after upload. After processing, the data can then be accessed by the patient via their software application and by a clinician via a dedicated clinician dashboard **1340**. The clinician may choose to activate a treatment protocol and remotely deliver the treatment signal through this wireless connectivity infrastructure.

[0083] Devices such as implants described herein are described as including communication devices and related electronics. These communications device(s) can include one or more transmitters and/or receivers (e.g., wireless and/or hard-wired transmitters/receivers). In various implementations, the communication devices are configured for a plurality of communication protocols, e.g., wireless protocols such as Wi-Fi, Bluetooth, BLE, Zigbee, etc., as well as radio communication and intercom communications, and/or a hardwired connection (e.g., fiber optic connection).

[0084] As noted herein, the implants and related electrodes disclosed according to various implementations provide numerous benefits relative to conventional implant apparatuses and systems. For example, various implementations can reduce complexity and/or invasiveness of conventional surgical procedures, e.g., by attaching the electrodes on or within the surface of the implant form. Further, relative to conventional implants, the disclosed implementations efficiently allocate electrodes and control modules, and also provide one or more separate components that can be assembled to traditional implant components within the surgical theater. Various implementations include assemblies, implants, systems and methods of coupling electrodes to smart orthopedic implant systems. Relative to conventional implants, the disclosed implants, assemblies and systems can improve patient outcomes and ease monitoring and/or treatment of patients post-operatively.

[0085] The functionality described herein, or portions thereof, and its various modifications (hereinafter “the functions”) can be implemented, at least in part, via a computer program product, e.g., a computer program tangibly embodied in an information carrier, such as one or more non-transitory machine-readable media, for execution by, or to control the operation of, one or more data processing apparatus, e.g., a programmable processor, a computer, multiple computers, and/or programmable logic components.

[0086] The term “approximately” as used with respect to values herein can allot for a nominal variation from absolute values, e.g., of several percent or less. Where the term “comprising” is used in the present description and claims, it does not exclude other elements or operations. The term “based on” (as in “A is based on B”) is used to indicate any of its ordinary meanings, including the cases (i) “based on at least” (e.g., “A is based on at least B”) and, if appropriate in the particular context, (ii) “equal to” (e.g., “A is equal to B”). Similarly, the term “in response to” is used to indicate any of its ordinary meanings, including “in response to at least.”

[0087] A computer program can be written in any form of programming language, including compiled or interpreted languages, and it can be deployed in any form, including as a stand-alone program or as a module, component, subroutine, or other unit suitable for use in a computing environment. A computer program can be deployed to be executed on one computer or on multiple computers at one site or distributed across multiple sites and interconnected by a network.

[0088] Actions associated with implementing all or part of the functions can be performed by one or more programmable processors executing one or more computer programs to perform the functions of the calibration process. All or part of the functions can be implemented as special purpose logic circuitry, e.g., an FPGA and/or an ASIC (application-specific integrated circuit). Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors, and any one or more processors of any kind of digital computer. Generally, a processor will receive instructions and data from a read-only memory or a random access memory or both. Components of a computer include a processor for executing instructions and one or more memory devices for storing instructions and data.

[0089] In various implementations, components described as being “coupled” to one another can be joined along one or more interfaces. In some implementations, these interfaces can include junctions between distinct components, and in other cases, these interfaces can include a solidly and/or integrally formed interconnection. That is, in some cases, components that are “coupled” to one another can be simultaneously formed to define a single continuous member. However, in other implementations, these coupled components can be formed as separate members and be subsequently joined through known processes (e.g., soldering, fastening, ultrasonic welding, bonding). In various implementations, electronic components described as being “coupled” can be linked via conventional hard-wired and/or wireless means such that these electronic components can communicate data with one another. Additionally, sub-components within a given component can be considered to be linked via conventional pathways, which may not necessarily be illustrated.

[0090] While inventive features described herein have been described in terms of preferred embodiments for achieving the objectives, it will be appreciated by those skilled in the art that variations may be accomplished in view of these teachings without deviating from the spirit or scope of the invention. Also, while this invention has been described according to a preferred use in spinal applications, it will be appreciated that it may be applied to various other uses desiring surgical fixation, for example, the fixation of long bones.

[0091] A number of implementations have been described. Nevertheless, it will be understood that additional modifications may be made without departing from the scope of the inventive concepts described herein, and, accordingly, other implementations are within the scope of the following claims.

Claims

1. An electrode assembly comprising: a first housing containing at least one electrode; and a control module with a first connector for mating with a complementary connector on the first housing, the control module including a controller programmed to initiate transmission of at least one of a measurement pattern or a stimulation pattern to a patient with the at least one electrode, wherein the first housing and the control module include a biocompatible material, and when connected, are sized to complement an orthopedic implant.
2. The electrode assembly of claim 1, wherein the controller is programmed to initiate transmission of both the measurement pattern and the stimulation pattern to the patient, and wherein the at least one electrode is configured for use in both the measurement pattern and the stimulation pattern.
3. The electrode assembly of claim 1, wherein the first housing contains a plurality of electrodes.
4. The electrode assembly of claim 1, further comprising a second housing containing at least one additional electrode, wherein the control module includes a second connector for mating with a complementary connector on the second housing.
5. The electrode assembly of claim 1, wherein when the first housing and the control module are connected, the electrode assembly is pliable to complement at least one contour in the orthopedic implant.
6. The electrode assembly of claim 5, wherein the electrode assembly is sized to sit within a groove or slot in the orthopedic implant.
7. The electrode assembly of claim 6, wherein the first housing includes a flexible insulative section adjacent the at least one electrode.
8. The electrode assembly of claim 1, wherein the first housing includes a plurality of electrodes interposed between adjacent insulative sections, wherein each of the plurality of electrodes is selectable for at least one of: measurement based on the measurement pattern or stimulation based on the stimulation pattern.
9. The electrode assembly of claim 8, wherein the plurality of electrodes includes at least three electrodes, wherein according to the stimulation pattern: a first electrode is selected as an anode and a second electrode is selected as a cathode, and wherein according to the measurement pattern: a first electrode is selected as a working electrode, a second electrode is selected as a counter electrode, and a third electrode is selected as a reference electrode.
10. The electrode assembly of claim 9, wherein the controller is configured to change selection of one or more of the plurality of electrodes according to the stimulation pattern or the measurement pattern.
11. The electrode assembly of claim 1, wherein the first housing includes a flexible tubular body and wherein the at least one electrode is integrated in a wall of the flexible tubular body such that a primary axis of each electrode is parallel with a primary axis of the flexible tubular body.
12. The electrode assembly of claim 1, wherein the first housing includes a flexible tubular body with an internal channel, and wherein the at least one electrode is integrated in a wall of the flexible tubular body such that a primary axis of each electrode is off-axis relative to a primary axis of the flexible tubular body.
13. The electrode assembly of claim 1, wherein the control module includes a hermetically sealed housing constructed from an implantable metal or a biocompatible polymer, wherein the housing includes: the controller, memory, a multiplexer, a power system, and a communication system, wherein the power system enables wireless power transfer to the control module and wherein the communication system enables wireless communication with an external control device.
14. The electrode assembly of claim 1, wherein the first housing and the control module have a substantially arcuate outer dimension or a substantially angled outer dimension.
15. The electrode assembly of claim 1, wherein the orthopedic implant includes a knee implant, a

tibial insert, a hip implant, or a surgical nail.
16. The electrode assembly of claim 1, wherein the first housing and the control module are integrally assembled, or configured for assembly during a surgical procedure on the patient.
17. An orthopedic implant comprising: a body for securing in a patient, the body including a mating feature; and an electrode assembly coupled with the body at the mating feature, the electrode assembly including: a first housing containing at least one electrode; and a control module connected with the first housing, the control module including a controller programmed to initiate transmission of at least one of a measurement pattern or a stimulation pattern to a patient with the at least one electrode, wherein the electrode assembly complements the mating feature on the body.
18. The orthopedic implant of claim 17, wherein the first housing includes a plurality of electrodes interposed between adjacent insulative sections, wherein each of the plurality of electrodes is selectable for at least one of: measurement based on the measurement pattern or stimulation based on the stimulation pattern.
19. The orthopedic implant of claim 17, wherein the controller is programmed to initiate transmission of both the measurement pattern and the stimulation pattern to a patient, and wherein the at least one electrode is configured for use in both the measurement pattern and the stimulation pattern.
20. The orthopedic implant of claim 17, wherein the mating feature includes a groove or a slot in the body, and wherein the electrode assembly is retained in the groove or slot in the body.